

Tidal dissipation within large icy satellites: Applications to Europa and Titan

G. Tobie*, A. Mocquet, C. Sotin

UMR 6112 Planétologie et Géodynamique, UFR des Sciences et Techniques, 2, rue de la Houssinière, BP 92208, 44322 Nantes cedex 03, France

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Abstract

This paper describes a new approach based on variational principles to calculate the radial distribution of tidal energy dissipation in any satellite. The advantage of the model with respect to classical solutions, is that it relates in a straightforward way the radial distribution of the time-averaged dissipation rate to its sensitivity to the corresponding distribution of viscoelastic parameters. This method is applied to Io-, Europa-, and Titan-like interiors, and it is tested against the results obtained by two classical methods by determining global dissipation as well as radial and lateral distributions within satellite interiors. By exploring systematically the different parameters defining the interior models, we demonstrate that the presence of a deep ocean below an outer ice layer strongly influences the tidal dissipation distribution in both the outer ice layer and in the innermost part of the satellite. On the one hand, the ocean by imposing a large radial displacement at the base of the outer ice layer, controls the distribution of tidal strain rate within the outer layer, making the tidal strain rate field very weakly sensitive to the viscosity variations. Conversely, in the high-pressure ice layer below the ocean, both tidal strain rate and dissipation are very sensitive to any variation of the ice viscosity. On the other hand, for identical structures of the mantle and of the core, the presence of a subsurface ocean reduces the strength of dissipation in the silicate mantle. The existence of a liquid layer within Europa makes models of the silicate mantle less dissipative than the predictions for Io.

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1. Introduction

The tremendous surface heat flux [$\geq 2 \text{ W m}^{-2}$ (Veeder et al., 1994; Matson et al., 2001; Rathbun et al., 2004)] and the intense volcanism of Io witness that tidal dissipation constitutes a significant heat source inside some satellites, and that it can strongly influence their thermal state (e.g., Segatz et al., 1988). In Europa, although tidal heating is probably lower due to the longer orbital period and the smaller radius of the satellite, it is still expected to contribute to the energy balance, and to favor the existence of a deep ocean below an ice shell thinner than 30 km

(Ojakangas and Stevenson, 1989; Hussmann et al., 2002; Tobie et al., 2003). In addition, tidal forces would be responsible for the complex surface activities of Europa (e.g., Greenberg et al., 1998). The observations of the complex tectonic structures on Europa's surface (e.g., Pappalardo et al., 1999) as well as the magnetic field data gathered by the Galileo probe (Kivelson et al., 2000) suggest that an ocean of liquid water is still present between a few kilometers and a few tens of kilometers beneath the cold icy surface. Magnetic data returned during the flybys of both Ganymede and Callisto may also indicate a conductive layer, probably made of salted water, at a depth of a few hundreds of kilometers within these two satellites (Zimmer et al., 2000; Kivelson et al., 2002).

The major constraint of the internal structure of the Galilean satellites is provided by the values of the mass M and of the MoI-factor, C/MR_s^2 , of the satellite, where C is

* Corresponding author. Fax : +33 (0) 2 51 12 52 68.
E-mail addresses: tobie@chimie.univ-nantes.fr (G. Tobie),
mocquet@chimie.univ-nantes.fr (A. Mocquet),
sotin@chimie.univ-nantes.fr (C. Sotin).

Table 1
Physical and orbital parameters of the Galilean satellites and of Titan

Satellite	Spherical radius R_s (km)	Mean density $\bar{\rho}$ (kg m ⁻³)	Mol-factor C/MR_s^2	Eccentricity e	Orbital period P_{orb} (days)	Angular frequency $\omega = 2\pi/P_{\text{orb}}$ (rad s ⁻¹)
Io	1821	3530	0.377	0.004	1.769	4.11×10^{-5}
Europa	1565	3010	0.346	0.01	3.552	2.05×10^{-5}
Ganymede	2634	1942	0.310	0.0015	7.154	1.02×10^{-5}
Callisto	2403	1830	0.355	0.007	16.689	4.36×10^{-6}
Titan	2575	1881	–	0.029	15.945	4.56×10^{-6}

Note. C is the principal moment of inertia, M is the mass of the satellite.

its principal moment of inertia (Mol) and R_s is its spherical radius, deduced from *Galileo* gravimetric data (Anderson et al., 1996, 1998, 2001) (see Table 1). The combination of these data with our knowledge of the Earth's interior and ice phase diagrams lead several authors to propose models for the interior of the icy satellites (e.g., Kuskov and Kronrod, 2001; Sohl et al., 2002; Sotin and Tobie, 2004). Following this, Io is exclusively made of rock and iron, whereas Europa, Ganymede, and Callisto are overlaid by a H₂O layer one to more than five hundred kilometers thick. Callisto is probably only partially differentiated, whereas the deep interior of the three other satellites may be differentiated into an iron core and a silicate mantle. The outermost H₂O shells of Ganymede and Callisto are expected to be thicker than the one of Europa (e.g., Sotin and Tobie, 2004). They are subdivided by pressure-induced phase transitions into ice I, ice III, ice V, and ice VI layers. The liquid layer, if it exists, would be present between the ice I layer and the ice V or ice VI layer (Grasset and Sotin, 1996). For Titan, the largest moon of Saturn, no similar data are yet available. By analogy with the jovian icy satellites, only theoretical models can give some constraints on its interior. Thermal evolution models (Lunine and Stevenson, 1987; Grasset et al., 2000) predict a differentiated structure close to Ganymede's one. The current internal structure of Titan significantly depends on the nature of the primitive materials that built it, and it remains quite unconstrained (Grasset et al., 2000). The dense atmosphere of Titan suggests the presence of volatiles, such as ammonia, within its interior (Lunine and Stevenson, 1987). This makes the existence of a liquid water layer very likely beneath several tens of kilometers of ice I (Grasset and Sotin, 1996; Grasset et al., 2000; Sohl et al., 2003).

During their travel around their giant planet on an eccentric orbit, the satellites are subjected to periodical deformations created by the tide-raising potential of the giant planet. A significant part of the tidal strain energy can be converted into heat by viscous friction within the viscoelastic interior or friction along faults at the surface. Although the latter mechanism in some circumstances could be locally a significant source of energy (e.g., Nimmo and Gaidos, 2002), it is unlikely that it affects the global dissipation of the satellite, and is then neglected in the present paper. The efficiency of the conversion by viscous friction depends on the thermal state of the interior. It is usually referred to as the specific

dissipation function Q^{-1} , or quality factor Q (e.g., Zschau, 1978). The amplitude of the tidal deformation and the associated energy are determined by the tide-raising potential Φ and by the internal structure of the satellite. For a synchronous eccentric orbit, the time-varying potential, to first order in the eccentricity, is given by (Kaula, 1964)

$$\Phi = r^2 \omega^2 e \left\{ -\frac{3}{2} P_2^0(\cos \theta) \cos \omega t + \frac{1}{4} P_2^2(\cos \theta) \times [3 \cos \omega t \cos 2\phi + 4 \sin \omega t \sin 2\phi] \right\}, \quad (1)$$

where r is the radius from the center of mass of the satellite, ω is the orbital angular frequency, e is the orbital eccentricity, θ and ϕ are the colatitude and longitude with zero longitude at the sub-jovian point, t is the time, and P_2^0 and P_2^2 are associated Legendre polynomials.

Tidal dissipation tends to make the orbit of the satellites circular and their rotation synchronous with their revolution. However, the orbital resonance of the Galilean satellites (Io, Europa, and Ganymede), known as the Laplace resonance, forces Io's and Europa's orbits to be eccentric, which allows for the long-term existence of significant eccentricities in spite of the strong tidal dissipation. Due to the circularization effect, the high eccentricity of Titan (~ 0.03) remains mysterious, since no significant resonance exists between Titan and any other satellite of Saturn to prevent the irremediable decay of its eccentricity since its accretion (Sohl et al., 1995). This requirement implies that tidal dissipation in Titan must be low compared to Io's and Europa's. However, even if it is lower, it may contribute significantly to the internal energy balance, and may help to maintain a liquid layer below several tens of kilometers of ice. Given their large semi-major axes, the small orbital eccentricities of Ganymede and Callisto indicate that tidal dissipation within their interiors can be safely neglected. Nevertheless, it is conceivable that Ganymede experienced higher dissipation in the past, when it came into the Laplace resonance (Showman and Malhotra, 1997). These authors proposed that the later phenomenon could partly explain the dichotomy between Callisto and Ganymede.

Classically, the global tidal dissipated power is computed from the imaginary part of the secondary-degree potential Love number k_2 (Zschau, 1978; Segatz et al., 1988)

$$\dot{E}_{\text{diss}} = -\frac{21}{2} \text{Im}(k_2) \frac{(\omega R_s)^5}{G} e^2, \quad (2)$$

where G is the gravitational constant. Though tidal heating is heterogeneously distributed (Segatz et al., 1988; Ojakangas and Stevenson, 1989; Tobie et al., 2003), the latter source of energy is commonly incorporated in thermal evolution models as uniform internal heating from the global value [Eq. (2)] (e.g., Peale et al., 1979; Fischer and Spohn, 1990; Showman et al., 1997; Hussmann et al., 2002; Moore, 2003; Sohl et al., 2003). The strong coupling between internal dynamics and tidal dissipation (Tackley et al., 2001; Tobie et al., 2003) requires the determination of both the radial and lateral distributions of the tidal heating, in order to correctly assess its effect on the thermal evolution.

For several decades, methods have been developed in the geophysical community to compute the free oscillations of the Earth as well as its response to both lunar and solar tides and to postglacial rebound (e.g., Takeushi and Saito, 1972; Wahr and Sasao, 1981; Dehant, 1987; Yuen et al., 1987; Lognonné, 1991; D'Agostino et al., 1997). In the present study, we propose a new method, derived from the classical theory of elastic body deformation and the energy density integrals commonly used in the seismological community, to compute the radial distribution of the tidal dissipation rate within the interior of any spherical body. The method is applied to a generic satellite, whose physical properties are close to those of the Galilean satellites and Titan. The physical and rheological parameters of the generic satellite are explored over a wide range of values, in order to isolate the key parameters of tidal dissipation. In Section 2, we briefly recall the classical formulation of the elastic body tide deformation, which is extended to the viscoelastic problem in Section 3. After a rapid description of classical concepts on tidal dissipation, Section 3 presents a new method based on the notion of the sensitivity of the time-averaged dissipation rate to the radial distribution of the rigidity. The method is applied on a generic satellite in Section 4. Finally, we discuss the implication of these calculations for Io, Europa, and Titan with a particular emphasis on the assessment of the tidal dissipation rate within icy layers.

2. Elastic formulation of body tide deformations

Tidal forcing by external bodies induces spheroidal modes of deformation inside and at the outer boundary of the planets and satellites. The instantaneous elastic spheroidal deformations of a spherically symmetric and non-rotating model of the latter bodies can be formulated by using six radial functions, y_i that satisfy a set of differential equations (Alterman et al., 1959; Takeushi and Saito, 1972; Jobert, 1973)

$$\frac{dy_i(r, \omega_l^m)}{dr} = \sum_{j=1}^6 \mathbf{A}_{ij} y_j(r, \omega_l^m), \quad (3)$$

where $y_i(r, \omega_l^m)$ is a function of radial distance r and angular frequency ω_l^m . Each frequency is associated with a

degree l and an azimuthal order m . The matrix $\mathbf{A}_{ij}$ depends on the angular frequency ω_l^m , on the radial distribution of the density ρ , bulk modulus K_E , and shear modulus μ_E in the isotropic elastic case, and on the acceleration due to gravity g . The elements of the matrix are defined by Eq. (82) of Takeushi and Saito (1972) for a transversely isotropic medium. The radial functions are associated with the radial and tangential displacements (y_1 and y_3 , respectively), the radial and tangential stresses (y_2 and y_4), and the gravitational potential (y_5). The sixth radial function (y_6) is defined such that the continuity of the gradient of the gravitational potential is taken into account. It is chosen in such a way that it simplifies the surface boundary condition

$$y_6(r, \omega_l^m) = \frac{dy_5(r, \omega_l^m)}{dr} - 4\pi G \rho y_1(r, \omega_l^m) + \frac{l+1}{r} y_5(r, \omega_l^m). \quad (4)$$

Note that the y_6 definition of Takeushi and Saito (1972) is different from the one proposed by Alterman et al. (1959). In spherical coordinates r , θ , and ϕ , the displacement $\vec{u} = (u_r, u_\theta, u_\phi)$ is expressed by

$$\begin{aligned} u_r(r, \theta, \phi) &= \sum_{l,m} y_1(r, \omega_l^m) \Phi_l^m(\theta, \phi), \\ u_\theta(r, \theta, \phi) &= \sum_{l,m} y_3(r, \omega_l^m) \frac{\partial \Phi_l^m(\theta, \phi)}{\partial \theta}, \\ u_\phi(r, \theta, \phi) &= \sum_{l,m} y_3(r, \omega_l^m) \frac{1}{\sin \theta} \frac{\partial \Phi_l^m(\theta, \phi)}{\partial \phi}, \end{aligned} \quad (5)$$

and the total gravitational potential Ψ , induced by a tide-generating potential $\Phi(\theta, \phi) = \sum_{l,m} \Phi_l^m(\theta, \phi)$, by

$$\Psi(r, \theta, \phi) = \sum_{l,m} y_5(r, \omega_l^m) \Phi_l^m(\theta, \phi). \quad (6)$$

2.1. Boundary conditions

At the surface, the radial and tangential stresses are equal to zero, and the gravitational potential must be continuous. As r approaches zero, it is required that the displacement and the disturbance potential also approach zero. The boundary conditions are:

$$y_1(0) = 0, \quad y_3(0) = 0, \quad y_5(0) = 0, \quad (7)$$

at the center of the body, and

$$y_2(R_s) = 0, \quad y_4(R_s) = 0, \quad y_6(R_s) = \frac{2l+1}{R_s}, \quad (8)$$

at the surface of the body. All the radial functions are continuous at internal boundaries, except at the interfaces between liquid and solid layers. We refer the reader to Takeushi and Saito (1972, pp. 253–255) for the boundary conditions at liquid-solid interfaces. The system of Eqs. (3) subject to the boundary conditions (7) and (8) is solved using a fourth order Runge–Kutta integration scheme.

2.2. Love numbers

At the surface, the so-called Love number, h_l , k_l , ℓ_l are defined by the radial functions y_i (e.g., [Jobert, 1973](#))

$$\begin{cases} h_l(\omega_l^m) = y_1(R_s, \omega_l^m)g(R_s), \\ k_l(\omega_l^m) = y_5(R_s, \omega_l^m) - 1, \\ \ell_l(\omega_l^m) = y_3(R_s, \omega_l^m)g(R_s). \end{cases} \quad (9)$$

The number h characterizes the radial displacement, k the gravitational potential induced by the mass redistribution, and ℓ the tangential displacement for a tide-generating potential equal to unity at the surface of the body.

2.3. Stress and strain tensors

From the six radial functions y_i , we compute the strain and stress tensors, at each grid point of the satellite. For elastic materials, each of the stress tensor components can be written as a linear function of all the strain tensor components. In spherical geometry, the strain tensor components associated with the displacement field are:

$$\begin{aligned} \varepsilon_{rr} &= \sum_{l,m} \frac{\partial y_1(r, \omega_l^m)}{\partial r} \Phi_l^m(\theta, \phi), \\ \varepsilon_{\theta\theta} &= \frac{1}{r} \sum_{l,m} y_3(r, \omega_l^m) \frac{\partial^2 \Phi_l^m(\theta, \phi)}{\partial \theta^2} + y_1(r, \omega_l^m) \Phi_l^m(\theta, \phi), \\ \varepsilon_{\theta\phi} &= \frac{2}{r} \sum_{l,m} y_3(r, \omega_l^m) \frac{1}{\sin \theta} \left[\frac{\partial^2 \Phi_l^m(\theta, \phi)}{\partial \phi \partial \theta} - \cot \theta \frac{\partial^2 \Phi_l^m(\theta, \phi)}{\partial \phi^2} \right], \\ \varepsilon_{\phi\phi} &= \frac{1}{r} \sum_{l,m} y_3(r, \omega_l^m) \frac{1}{\sin^2 \theta} \frac{\partial^2 \Phi_l^m(\theta, \phi)}{\partial \phi^2} \\ &\quad + [y_3(r, \omega_l^m) \cot \theta + y_1(r, \omega_l^m)] \Phi_l^m(\theta, \phi), \\ \varepsilon_{\phi r} &= \frac{1}{\mu_E} \sum_{l,m} y_4(r, \omega_l^m) \frac{1}{\sin \theta} \frac{\partial \Phi_l^m(\theta, \phi)}{\partial \phi}, \\ \varepsilon_{r\theta} &= \frac{1}{\mu_E} \sum_{l,m} y_4(r, \omega_l^m) \frac{\partial \Phi_l^m(\theta, \phi)}{\partial \theta}. \end{aligned} \quad (10)$$

In the following paragraphs, the radial and frequency dependences of the variables and parameters will not be written explicitly unless it is necessary. Similarly, for sake of clarity ω will implicitly mean ω_l^m .

3. Viscoelastic response and tidal dissipation

3.1. Viscoelastic rheology and correspondence principle

In practice, at typical tidal forcing periods (a few days for the Galilean satellites and Titan), the planetary interiors

behave like viscoelastic bodies. This implies that the equilibrium strain of the disturbed body is delayed with respect to the onset time of forcing. A key parameter of the viscoelastic response is the Maxwell time τ_M that is defined as the ratio between the effective Newtonian viscosity η and the elastic shear modulus μ_E of the material: $\tau_M = \eta/\mu_E$. Although creep mechanisms in ice at low stress levels, typical of tidal forcing ($\sigma \simeq 0.01$ – 0.1 MPa) remain uncertain and strongly debated (see [Montagnat and Duval, 2000](#); [Durham et al., 2001](#); [Goldsby and Kohlstedt, 2001](#); [Duval and Montagnat, 2002](#); [Goldsby and Kohlstedt, 2002](#)), most of mechanical tests and polar ice sheet analysis agree with a quasi-Newtonian behavior of ice Ih at very low stresses (for a review see Table 4 in [Goldsby and Kohlstedt, 2001](#)) and with an effective Newtonian viscosity of around 10^{14} Pa s near the melting point [see [Sotin and Tobie \(2004\)](#) for a more complete discussion]. Then, at expected values of shear modulus ($\sim 3 \times 10^9$ Pa) for ice Ih, the Maxwell time is close to the satellite orbital period.

The latter estimates lead us to assume a compressible Maxwell rheology, described by three parameters: the elastic shear modulus μ_E , the incompressibility K_E and the Newtonian viscosity η . Dissipation is thus only considered in shear motions. Nevertheless, all the theoretical developments presented here are not restricted to this assumption and can be applied to any other rheological model. In the Maxwell model, the stress-strain relation is written in Cartesian coordinates

$$\begin{aligned} \dot{\sigma}_{ij} + (\mu_E/\eta) \left(\sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij} \right) \\ = 2\mu_E \dot{\varepsilon}_{ij} + \left(K_E - \frac{2}{3} \mu_E \right) \dot{\varepsilon}_{kk} \delta_{ij}, \end{aligned} \quad (11)$$

where δ_{ij} is Kronecker's symbol. In Eq. (11), summation over repeated indices is implicit. This classical convention is used in the rest of the paper, otherwise stated differently.

Using the correspondence principle established by [Biot \(1954\)](#), the elastic problem formulation stated in Section 2 can be extended to the resolution of an equivalent viscoelastic problem. Provided zero initial conditions and similar geometries in the elastic and viscoelastic cases, the Laplace and Fourier transformed viscoelastic equations, and the boundary conditions, are formally identical to the elastic equations when the rheological parameters and the radial functions are defined by complex values. The Fourier transform of Eq. (11) leads to a Hooke-like law

$$\tilde{\sigma}_{ij} = 2\tilde{\mu}(\omega) \tilde{\varepsilon}_{ij} + \left[K_E - \frac{2}{3} \tilde{\mu}(\omega) \right] \tilde{\varepsilon}_{kk} \delta_{ij}, \quad (12)$$

where

$$\tilde{\mu}(\omega) = \frac{\mu_E \omega^2 \eta^2}{\mu_E^2 + \omega^2 \eta^2} + i \frac{\mu_E^2 \omega \eta}{\mu_E^2 + \omega^2 \eta^2}, \quad (13)$$

and the tilde indicates complex values in the frequency domain. For tidal forcing periods of a few days, we can reasonably assume that the response frequency of the body is equal

to the forcing frequency ω_f . In other words, coupled modes, which are usually encountered in free oscillation problems at seismic frequencies (e.g., Lognonné, 1991) are not considered here.

The application of the correspondence principle consists in replacing the radial functions y_i , the parameter μ_E and the Love numbers (e.g., k_l) in the set of Eqs. (3)–(10) by their complex Fourier transforms, $\tilde{y}_i$, $\tilde{\mu}$, and $\tilde{k}_l$, respectively. In the following paragraphs, for sake of writing simplicity, the tilde will be removed and the complex values will be noted y_i , μ , and k_l . The real parts of these complex numbers correspond to the instantaneous elastic response; the imaginary parts govern the dissipation of energy by internal friction, and thus the delay of the response with respect to the tide-raising potential (e.g., Zschau, 1978; Dehant, 1987). Note that the new method further described and used in the present paper, can be applied for any other rheological model by redefining Eqs. (11) to (13) with respect to the rheological assumptions.

3.2. Calculations of tidal dissipation

Though the concept of energy balance is detailed in many textbooks (e.g., Aki and Richards, 1980; Ranalli, 1995), we recall here some definitions that will be useful afterwards. After deriving the general approach for computing the dissipation rate per unit volume, we detail a new method based on the application of variational principles to the radial functions. This method enables us to calculate the radial distribution of the tidal energy dissipation. Its main advantage with respect to classical solutions (e.g., Segatz et al., 1988), is that it relates in a straightforward way the radial distribution of the time-averaged dissipation rate to its sensitivity to the corresponding distribution of viscoelastic parameters.

3.2.1. Energy balance

The first law of thermodynamics states that any body possesses an internal energy, which varies with its deformation. The rate of doing mechanical work $\dot{E}_{\text{mech}}$ plus the rate of dissipation $\dot{E}_{\text{diss}}$ are equal to the rate of increase of the kinetic and internal energies of the body, $\dot{E}_{\text{kin}}$ and $\dot{E}_{\text{int}}$, respectively,

$$\dot{E}_{\text{mech}} + \dot{E}_{\text{diss}} = \dot{E}_{\text{kin}} + \dot{E}_{\text{int}}. \quad (14)$$

The rate of doing mechanical work $\dot{E}_{\text{mech}}$ is given by

$$\dot{E}_{\text{mech}} = \iiint_V \rho \frac{\partial \vec{u}}{\partial t} \cdot \text{grad}(\Psi) dV + \iint_S \frac{\partial \vec{u}}{\partial t} \cdot \vec{\tau}_s dS, \quad (15)$$

where $\vec{\tau}_s$ is the surface traction, V and S are the volume and the outer surface of the deformed body, respectively. Using the equation of motion and Gauss's divergence theorem, Eq. (15) can be written

$$\dot{E}_{\text{mech}} = \frac{\partial}{\partial t} \iiint_V \frac{1}{2} \rho \dot{u}_i^2 dV + \iiint_V \sigma_{ij} \dot{\epsilon}_{ij} dV. \quad (16)$$

Assuming that all the dissipated energy is converted into heat, the rate of heating $\dot{E}_{\text{diss}}$ is given by

$$\dot{E}_{\text{diss}} = \iiint_V h dV, \quad (17)$$

where h is the heating rate per unit volume, and the rate of kinetic energy $\dot{E}_{\text{kin}}$ is defined

$$\dot{E}_{\text{kin}} = \frac{\partial}{\partial t} \iiint_V \frac{1}{2} \rho \dot{u}_i^2 dV. \quad (18)$$

Thus, the rate of the increase of internal energy per unit volume $\dot{U}$ is expressed as

$$\dot{U} = h + \sigma_{ij} \dot{\epsilon}_{ij}. \quad (19)$$

For a purely elastic deformation, the heat content of any unit volume remains constant, and

$$\dot{U}_{\text{elas}} = \sigma_{ij} \dot{\epsilon}_{ij}. \quad (20)$$

For a viscoelastic deformation, $h > 0$.

3.2.2. Dissipation rate per unit volume

By using the Fourier transformation of the stress and strain rate tensors, and assuming that the body responds at the frequency ω_f imposed by the forcing, the stress tensor σ_{ij} and the strain rate tensor $\dot{\epsilon}_{ij}$ are written as a function of time

$$\sigma_{ij}(t) = \int_0^\infty \tilde{\sigma}_{ij}(\omega) e^{i\omega t} \delta(\omega - \omega_f) d\omega = \tilde{\sigma}_{ij}(\omega_f) e^{i\omega_f t}, \quad (21)$$

$$\begin{aligned} \dot{\epsilon}_{ij}(t) &= \frac{\partial}{\partial t} \int_0^\infty \tilde{\epsilon}_{ij}(\omega) e^{i\omega t} \delta(\omega - \omega_f) d\omega \\ &= i\omega_f \tilde{\epsilon}_{ij}(\omega_f) e^{i\omega_f t}, \end{aligned} \quad (22)$$

where $\tilde{\sigma}_{ij}(\omega_f)$ and $\tilde{\epsilon}_{ij}(\omega_f)$ are the complex amplitudes of the stress and strain tensors, verifying Eq. (12) in the case of a compressible Maxwell body. Their values are calculated from the complex radial functions y_i .

A complex power P^c per unit volume is defined from the complex components of the stress and strain tensors

$$\begin{aligned} 2P^c &= i\omega_f \tilde{\sigma}_{ij} \tilde{\epsilon}_{ij}^* = \omega_f \underbrace{[\text{Im}(\tilde{\sigma}_{ij}) \text{Re}(\tilde{\epsilon}_{ij}) - \text{Re}(\tilde{\sigma}_{ij}) \text{Im}(\tilde{\epsilon}_{ij})]}_{\text{real}} \\ &\quad - i\omega_f \underbrace{[\text{Re}(\tilde{\sigma}_{ij}) \text{Re}(\tilde{\epsilon}_{ij}) + \text{Im}(\tilde{\sigma}_{ij}) \text{Im}(\tilde{\epsilon}_{ij})]}_{\text{imaginary}}, \end{aligned} \quad (23)$$

the star indicating a complex conjugate. The real part of the complex power represents the dissipated power averaged over one forcing cycle, whereas the imaginary part corresponds to the elastic energy instantaneously stored by the body deformation during one quarter of cycle and restored during the next quarter. The tidal heating rate per unit volume averaged over one cycle is then equal to

$$\bar{h}_{\text{tide}} = \frac{\omega_f}{2} [\text{Im}(\tilde{\sigma}_{ij}) \text{Re}(\tilde{\epsilon}_{ij}) - \text{Re}(\tilde{\sigma}_{ij}) \text{Im}(\tilde{\epsilon}_{ij})]. \quad (24)$$

Table 2

Expression of the energy integrals I_i in their general bilinear forms and in the particular case where the x_i function is the conjugate of the complex function y_i

	x_i any complex function	$x_i = y_i^*$
I_1	$\int_0^{R_s} \rho [y_1 x_1 + l(l+1) y_3 x_3] r^2 dr$	$\int_0^{R_s} \rho [y_1 ^2 + l(l+1) y_3 ^2] r^2 dr$
I_2	$\int_0^{R_s} \sum_{k=1}^9 I_{2,k} dr$	$\int_0^{R_s} \sum_{k=1}^9 I_{2,k} dr$
I_{21}	$(K + \frac{4}{3}\mu) r^2 \frac{dx_1}{dr} \frac{dy_1}{dr}$	$\frac{4}{3} \frac{r^2}{ K+4/3\mu ^2} y_2 - \frac{K-2/3\mu}{r} [2y_1 - l(l+1)y_3] ^2 \times \mu$ $+ \frac{r^2}{ K+4/3\mu ^2} y_2 - \frac{K-2/3\mu}{r} [2y_1 - l(l+1)y_3] ^2 \times K$
I_{22}	$(K - \frac{2}{3}\mu) r (\frac{dx_1}{dr} [2y_1 - l(l+1)y_3] + \frac{dy_1}{dr} [2x_1 - l(l+1)x_3])$	$-\frac{4}{3} r \operatorname{Re} \{ \frac{dy_1^*}{dr} [2y_1 - l(l+1)y_3] \} \times \mu$ $+ 2r \operatorname{Re} \{ \frac{dy_1^*}{dr} [2y_1 - l(l+1)y_3] \} \times K$
I_{23}	$(K + \frac{1}{3}\mu) [2y_1 - l(l+1)y_3] [2x_1 - l(l+1)x_3]$	$\frac{1}{3} 2y_1 - l(l+1)y_3 ^2 \times \mu + 2y_1 - l(l+1)y_3 ^2 \times K$
I_{24}	$l(l+1) r y_4 (r \frac{dx_3}{dr} + x_1 - x_3)$	$l(l+1) r^2 y_4 ^2 / \mu ^2 \times \mu$
I_{25}	$l(l^2 - 1)(l+2) \mu x_3 y_3$	$l(l^2 - 1)(l+2) y_3 ^2 \times \mu$
I_{26}	$(l+1) \rho r (x_1 y_5 + x_5 y_1)$	$2(l+1) r \operatorname{Re} \{ y_1^* y_5 \} \times \rho$
I_{27}	$-l(l+1) \rho r (x_3 y_5 + x_5 y_3)$	$-2l(l+1) r \operatorname{Re} \{ y_3^* y_5 \} \times \rho$
I_{28}	$-\rho g r (x_1 [2y_1 - l(l+1)y_3] + y_1 [2x_1 - l(l+1)x_3])$	$-2g r \operatorname{Re} \{ y_1^* [2y_1 - l(l+1)y_3] \} \times \rho$
I_{29}	$(4\pi G)^{-1} r y_6 [r \frac{dx_5}{dr} - (4\pi G) \rho r x_1 + (l+1)x_5]$	$(4\pi G)^{-1} r y_6 ^2$
I_3	$\{ r^2 [x_1 y_2 + l(l+1) x_3 y_4 + \frac{x_5 y_6}{4\pi G}] \}_{r=R_s}$	$\{ r^2 [y_1^* y_2 + l(l+1) y_3^* y_4 + \frac{y_5^* y_6}{4\pi G}] \}_{r=R_s}$

3.3. Variational equations and tidal dissipation

Following the classical approach used in free oscillation problems (Backus and Gilbert, 1968; Takeushi and Saito, 1972; for a review, see Romanowicz and Durek, 2000), we propose to compute the radial distribution of the dissipation rate, using the variational principle. The application of the variational technique to the system of differential equations (3) provides explicit expressions of the kinetic and strain energy–density integrals associated with the motion. Considering a spherically symmetric model, we incorporate Eqs. (4)–(6) into Eqs. (15), (16), and (18). In the frequency domain, and using Green’s first transformation, the set of Eqs. (14)–(18) reduces to

$$\iiint_V \sigma_{ij} \dot{\epsilon}_{ij} dV - \iiint_V \rho \frac{\partial \vec{u}}{\partial t} \cdot \vec{\text{grad}}(\Psi) dV = \omega I_2 \int_0^\pi \int_0^{2\pi} [\Phi(\theta, \phi)]^2 \sin \theta d\theta d\phi, \quad (25)$$

$$\dot{E}_{\text{mech}} = \iiint_V \rho \frac{\partial \vec{u}}{\partial t} \cdot \vec{\text{grad}}(\Psi) dV + \omega I_3 \int_0^\pi \int_0^{2\pi} [\Phi(\theta, \phi)]^2 \sin \theta d\theta d\phi, \quad (26)$$

$$\dot{E}_{\text{kin}} = -\omega^3 I_1 \int_0^\pi \int_0^{2\pi} [\Phi(\theta, \phi)]^2 \sin \theta d\theta d\phi, \quad (27)$$

where I_1 , I_2 , and I_3 are the radial density integrals related to the kinetic, strain and potential energies, respectively (Takeushi and Saito, 1972). The conservation of energy [Eq. (14)] requires

$$I_3 = -\omega^2 I_1 + I_2. \quad (28)$$

For any set of arbitrary complex functions x_i ($i = 1, 2, \dots, 6$) that do not necessarily verify the boundary conditions of the problem, the general bilinear forms of the complex energy integrals I_1 , I_2 , and I_3 , are written (Takeushi and Saito, 1972)

$$I_1 = \int_0^{R_s} \rho [y_1 x_1 + l(l+1) y_3 x_3] r^2 dr, \quad (29)$$

$$I_2 = \int_0^{R_s} \sum_{k=1}^9 I_{2,k} dr, \quad (30)$$

$$I_3 = \left\{ r^2 \left[x_1 y_2 + l(l+1) x_3 y_4 + \frac{x_5 y_6}{4\pi G} \right] \right\}_{r=R_s}. \quad (31)$$

The terms $I_{2,k}$ are defined in Table 2.

If we choose the conjugate of the y_i functions for the x_i functions, the integrals can be written as the sum of a real term and of an imaginary one. The latter corresponds to the dissipation. Table 2 presents the expression of the different integrals. Rearranging the I_2 integrals as

$$I_2 = \int_0^{R_s} [H_K K + H_\mu \mu + H_\rho \rho + H_0] dr, \quad (32)$$

H_K , H_μ , and H_ρ represent the radial sensitivity to the bulk modulus K , to the shear modulus μ and to the density ρ , respectively. These parameters are purely real. They depend on the radial structure of the satellite and on the radial functions y_i :

$$\begin{aligned}
 H_K &= \frac{r^2}{|K + 4/3\mu|^2} \left| y_2 - \frac{K - 2/3\mu}{r} [2y_1 - l(l+1)y_3] \right|^2 \\
 &\quad + 2r \operatorname{Re} \left\{ \frac{dy_1^*}{dr} [2y_1 - l(l+1)y_3] \right\} \\
 &\quad + |2y_1 - l(l+1)y_3|^2, \\
 H_\mu &= \frac{4}{3} \frac{r^2}{|K + 4/3\mu|^2} \left| y_2 - \frac{K - 2/3\mu}{r} [2y_1 - l(l+1)y_3] \right|^2 \\
 &\quad - \frac{4}{3} r \operatorname{Re} \left\{ \frac{dy_1^*}{dr} [2y_1 - l(l+1)y_3] \right\} \\
 &\quad + \frac{1}{3} |2y_1 - l(l+1)y_3|^2 + l(l+1)r^2 |y_4|^2 / |\mu|^2 \\
 &\quad + l(l^2 - 1)(l+2) |y_3|^2, \\
 H_\rho &= 2(l+1)r \operatorname{Re} \{ y_1^* y_5 \} - 2l(l+1)r \operatorname{Re} \{ y_3^* y_5 \} \\
 &\quad - 2gr \operatorname{Re} \{ y_1^* [2y_1 - l(l+1)y_3] \}, \\
 H_0 &= (4\pi G)^{-1} r |y_6|^2.
 \end{aligned} \tag{33}$$

The bulk dissipation $\operatorname{Im}\{K\}$ of the planetary materials is very badly constrained and, a priori much smaller than $\operatorname{Im}\{\mu\}$. We thus assume that all the information about dissipation is contained in the imaginary part of the complex shear modulus $\mu(\omega)$. Using the surface boundary conditions (8), Eq. (31) becomes

$$I_3 = \frac{(2l+1)R_s}{4\pi G} y_5(R_s). \tag{34}$$

Equating the function x_i with the conjugate of the functions y_i , and introducing the expressions (29) and (32) into the energy balance (28), we obtain

$$\frac{(2l+1)R_s}{4\pi G} \operatorname{Im}\{y_5\}_{r=R_s} = \int_0^{R_s} H_\mu \operatorname{Im}\{\mu\} dr. \tag{35}$$

For $l = 2$, since $\operatorname{Im}\{y_5\}_{r=R_s} = \operatorname{Im}\{k_2\}$ [see Eq. (9)], and inserting $\operatorname{Im}\{y_5\}_{r=R_s}$ in Eq. (2), the global tidal dissipated power can thus be expressed as

$$\dot{E}_{\text{diss}} = -\frac{42\pi}{5} \omega^5 R_s^4 e^2 \int_0^{R_s} H_\mu \operatorname{Im}\{\mu\} dr, \tag{36}$$

and the radial distribution of the dissipation rate per unit volume is

$$h_{\text{tide}}(r) = -\frac{21}{10} \frac{\omega^5 R_s^4 e^2}{r^2} H_\mu \operatorname{Im}\{\mu\}. \tag{37}$$

Equation (37) enables us to compute the radial distribution of the dissipation rate within any planetary interior using the values of $\operatorname{Im}\{\mu\}$ and H_μ provided by Eqs. (13) and (33), respectively. Note that the latter variational approach does

not require any assumption regarding the value of the shear dissipation $\operatorname{Im}\{\mu\}$.

4. Tidal dissipation in a generic satellite

4.1. Definition of a generic satellite

In order to explore all physical parameters that may control the tidal response of the satellite, we define a generic satellite whose physical properties are close to those of the different satellites presented in Table 1.

4.1.1. Internal structure

For the rocky interior, we consider two different models: an undifferentiated homogeneous structure and another one differentiated into a metallic core and an olivine-rich mantle in agreement with models proposed by Sohl et al. (2002) for the three innermost Galilean satellites (see Fig. 1). The radius R_M of the rocky interior is fixed to 1600 km, which is almost the mean value of Europa's and Io's rocky part. The mean density $\bar{\rho}$ of the rocky interior is prescribed to 3500 kg m^{-3} , which corresponds to Io's mean density. The density ρ_M of the mantle is fixed to 3300 kg m^{-3} , a typical value for the Earth's upper mantle. Since core densities ρ_C may vary between roughly 5000 kg m^{-3} for a Fe/FeS core (Sanloup et al., 2000) to 8000 kg m^{-3} for a pure iron core, the radius R_C of the core-mantle boundary (CMB here after) is calculated for the mean density $\bar{\rho}$ to remain constant

$$R_C = \left[\frac{\bar{\rho} - \rho_M}{\rho_C - \rho_M} \right]^{1/3} R_M. \tag{38}$$

This interior is overlain by a more or less thick H_2O layer. Two different models are considered:

- An Europa-like model consisting of a thin H_2O layer. Its thickness is calculated such that the mean density of the satellite is equal to 3000 kg m^{-3} , Europa's mean density. The layer is divided in a liquid and a solid part, made

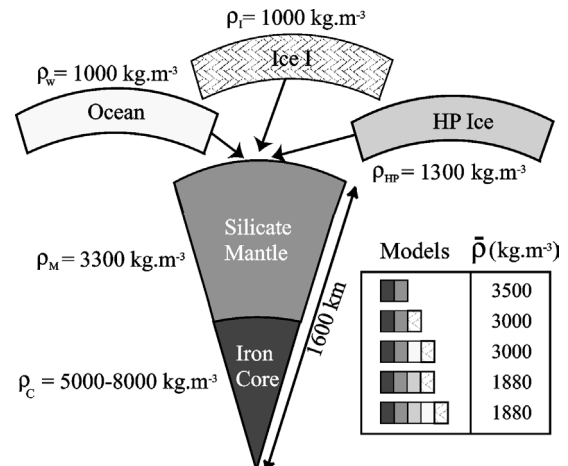


Fig. 1. Models of the generic satellite.

Table 3
Reference parameter values for each layer of the generic satellite

Layer	V_S (m s ⁻¹)	V_P (m s ⁻¹)	η (Pa.s)	References
Homogeneous interior	4500	8000	10^{20}	
Liquid Iron core	0	6000	0	
Silicate mantle	4500	8000	10^{20}	Dziewonski and Anderson (1981)
High-pressure ice	2000	4000	10^{14}	Sotin et al. (1998)
Ocean	0	1500	0	
Ice I	2000	4000	10^{14}	Sotin et al. (1998)

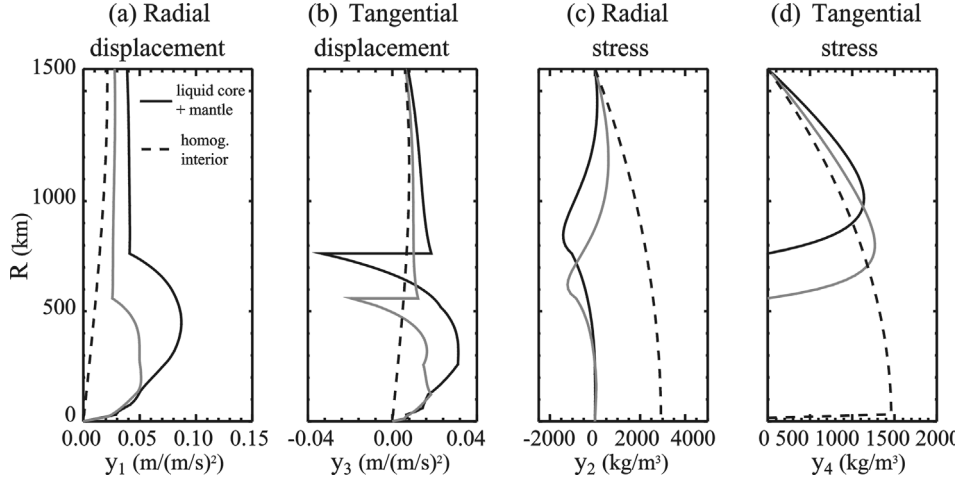


Fig. 2. Radial functions y_i within the rocky interior for an homogeneous undifferentiated model (dashed curve) and models differentiated into a silicate mantle and a liquid core with two different density values [5150 (black solid curves)/8000 (gray solid curves) kg m⁻³].

of ice I. For sake of simplicity, following Moore and Schubert (2000), we use the same density for the water layer and the ice I layer: $\rho_w = \rho_I = 1000$ kg m⁻³.

- A Titan-like model consisting of a thick H₂O layer, divided in a high-pressure sublayer, a liquid sublayer and an ice I one. In the high-pressure layer, we neglect the ice phase transition, and we assume a mean density ρ_{HP} of 1300 kg m⁻³, corresponding almost to the density of ice VI (Sotin et al., 1998). For the two other layers, the density values are equal to the values of the Europa-like model. The mean density of the satellite is prescribed to that of Titan: $\bar{\rho} = 1880$ kg m⁻³.

Note that these two models provide neither the exact radius of Europa and of Titan, nor the value of Europa's MoI. The goal of the following calculations is to understand the effect of each parameter on the distribution of the tidal dissipation rate, for different interior models, all starting from the same deep interior structure.

4.1.2. Parameter values for each internal layer

Several parameters are explored one after another over a wide range of values. If one of the parameters is variable, the other ones are fixed at the values listed in Table 3, where V_S and V_P are the values of the elastic S- and P-wave velocities, respectively. The latter velocities are linked to the values of

the elastic shear modulus, μ_E , and of the elastic bulk modulus, K_E by the following relationships:

$$\mu_E = \rho V_S^2, \quad (39)$$

$$K_E = \rho V_P^2 - \frac{4}{3}\mu_E. \quad (40)$$

4.2. Dissipation in the rocky interior

To start with, we limit our study to homogeneous and two-layer rocky models in order to isolate the effect of each parameter on the dissipation within the deep interior. We neglect dissipation in the liquid core by setting the value of the viscosity to zero.

4.2.1. Radial functions

Fig. 2 represents the radial functions obtained for an homogeneous undifferentiated model and for two-layer differentiated models. These functions control the global deformation as well as its distribution within the interior. The radial functions, y_1 , y_2 , y_3 , and y_4 , are proportional to the radial displacement, the radial stress, the tangential displacement, and the tangential stress, respectively. They are computed by solving the system of Eqs. (3) in the viscoelastic case, subject to the boundary conditions (7) and (8).

If the interior is undifferentiated (dashed curves in Fig. 2), it undergoes tensile (positive) stresses that decrease from the

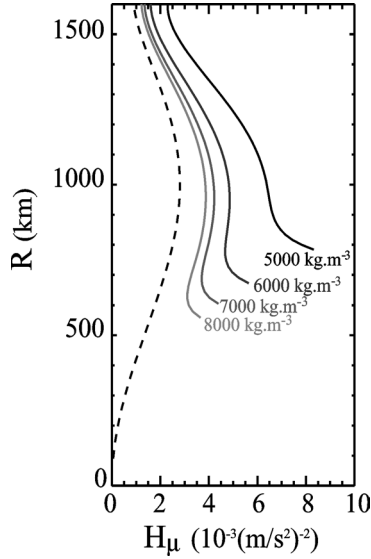


Fig. 3. Sensitivity parameter H_μ within the silicate mantle for different values of the core density (solid curves) and within an homogeneous undifferentiated interior (dashed curve).

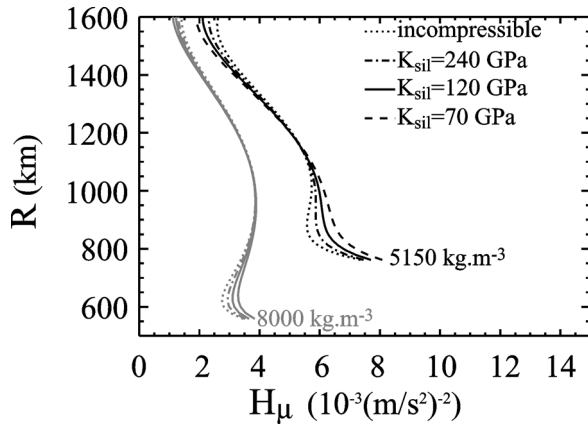


Fig. 4. Sensitivity parameter H_μ within the silicate mantle for different values of the silicate bulk modulus K_{sil} and two different core density values.

center to the surface (Fig. 2c), whereas, if the interior is differentiated into a liquid core and a silicate mantle (solid curves in Fig. 2), the base of the mantle undergoes com-

pressive (negative) radial stresses. The large displacement of the poorly compressible liquid core acts in pushing up the base of the mantle (Fig. 2a). The smaller displacement at the surface of the mantle than at its base, results in a radial compression of the mantle.

The state of the interior changes drastically the tangential components (Figs. 2b and 2d). The tangential stress vanishes in the liquid core, and the amplitude of the displacement is very much larger than in the undifferentiated interior. A liquid core induces large tangential displacements at the core-mantle boundary, and an increase of torsional stresses above the latter interface (Fig. 2d).

4.2.2. Radial distribution: sensitivity to shear anelasticity

Fig. 3 shows the radial distribution of the parameter H_μ , which represents the sensitivity of the tidal dissipation rate to shear anelasticity [see Eq. (37)], for two interior models and different values of the core density ρ_C . The distribution of the sensitivity parameter H_μ is mainly controlled by the state of the interior. The difference between the undifferentiated case and the differentiated case with a liquid core is particularly large when the core density is low. The liquid state of the core induces a strong increase of the shear modulus sensitivity toward the core-mantle boundary. This is due to the large contrast of the tangential displacement across the core-mantle boundary (see Fig. 2b).

The silicate bulk modulus K_{sil} also modifies the distribution of H_μ in the mantle as well as in the homogeneous interior to some extent. Fig. 4 illustrates this effect in the case of a differentiated interior. With decreasing values of the bulk modulus K_{sil} , i.e. if the compressibility of the mantle increases, the sensitivity parameter H_μ increases at the base of the mantle and decreases at its surface. These two opposite effects lead to an almost constant value for H_μ when it is averaged throughout the whole mantle, so that the global response of the satellite is insensitive to the value of the bulk modulus K_{sil} of the mantle.

Inversely, the values of the silicate elastic shear modulus μ_{sil} and viscosity η_{sil} do not influence the shape of the radial distribution of the sensitivity parameter H_μ , but modify its radially averaged value. As Fig. 5 illustrates in the case of a differentiated structure, the mean value of

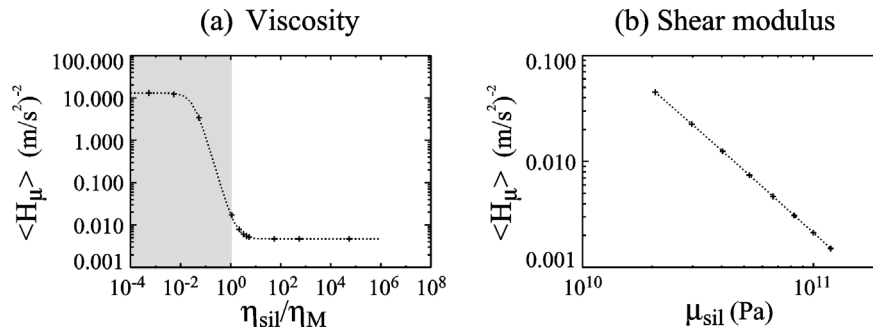


Fig. 5. Sensitivity parameter $\langle H_\mu \rangle$ averaged throughout the silicate mantle as a function of the silicate viscosity η_{sil} (a) and as a function of the silicate shear modulus μ_{sil} (b) within the mantle for a liquid core of 5150 kg m^{-3} . For the latter results, the ratio between the velocities V_P and V_S is prescribed to $\sqrt{3}$, such that the Lamé constant, λ_{sil} , is always equal to the shear modulus μ_{sil} . The gray area indicates the viscosity values that are physically unreliable.

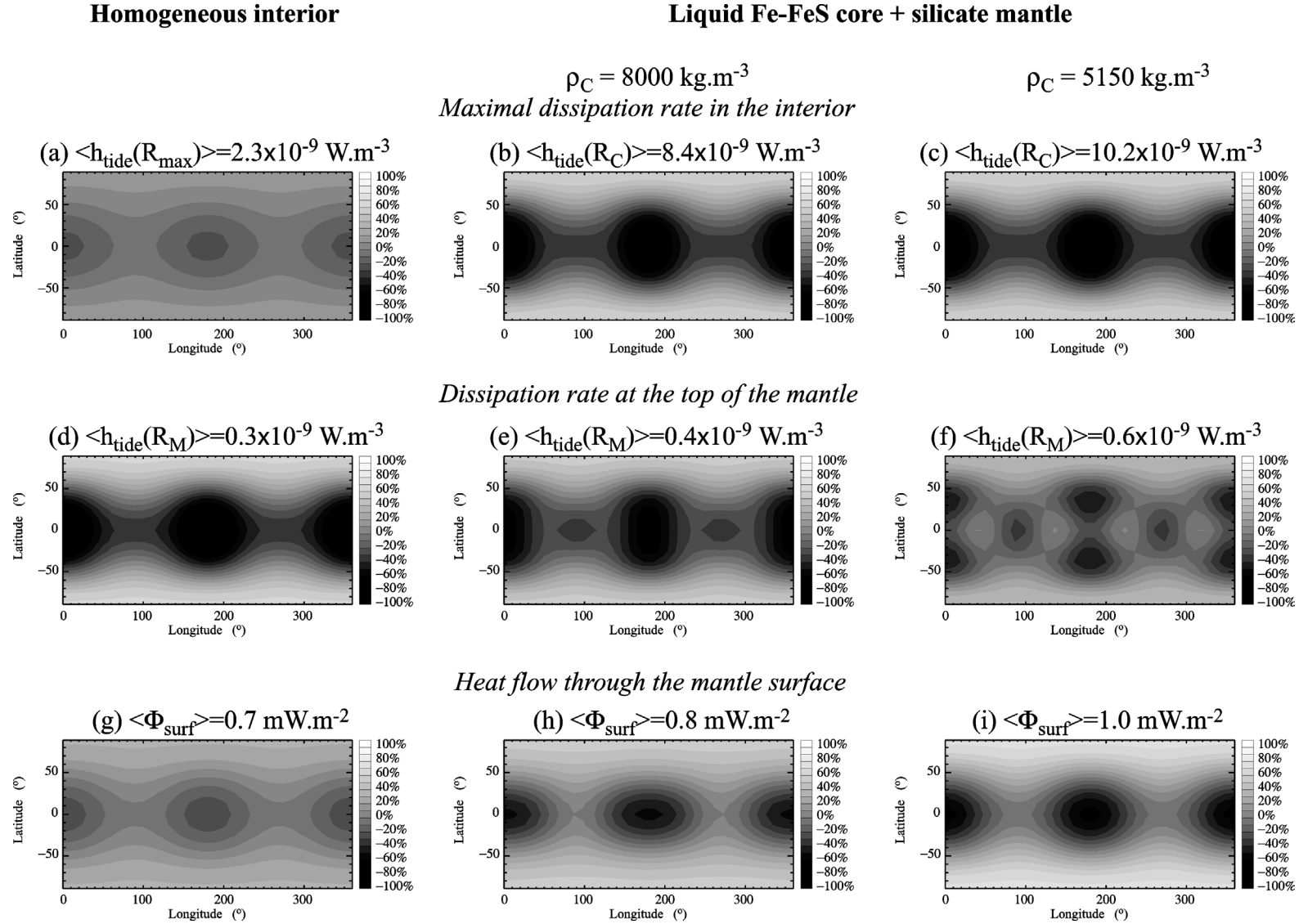


Fig. 6. Lateral distribution, in percent (%) of the mean value indicated above each panel, of the maximal tidal dissipation rate within the silicate interior (a, b, c), of the tidal dissipation rate at the interior surface (d, e, f), and of the surface heat flow (g, h, i) for an homogeneous interior model (left column) and differentiated models with two different core density values [8000 kg m^{-3} (middle column), 5150 kg m^{-3} (right column)].

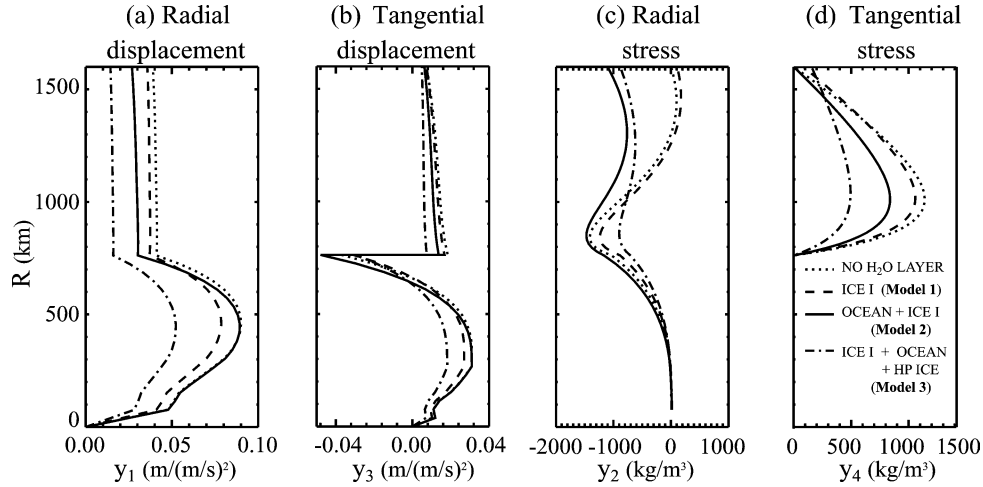


Fig. 7. Radial functions y_i in the rocky interior for different outer H₂O layer models, for a liquid core of 5150 kg m^{-3} .

H_μ linearly increases when the shear modulus decreases, and exponentially increases when the viscosity reaches the Maxwell viscosity value η_M . Below this value, the mantle tends to behave like a purely viscous material. Since the mean value of the mantle viscosity cannot be realistically smaller than 10^{17} Pa s , the ratio $(\eta_{\text{sil}}/\eta_M)$ is likely to be larger than 10^2 . Thus, for realistic values of silicate viscosity ($\eta_{\text{sil}}/\eta_M \geq 10^2$), the H_μ parameter is not sensitive to viscosity values, and it is mainly controlled by the elastic shear modulus value. Additional test performed for a wide range of values have shown that the bulk modulus of the liquid core as well as the orbital frequency have no significant influence either on the shape of the H_μ profile or on the mean value of H_μ .

4.2.3. Lateral distribution

The lateral distribution of the tidal dissipation rate within the interior is computed from the strain and stress tensors using Eq. (24). For the following calculations, the silicate viscosity η_{sil} is prescribed to 10^{18} Pa s , the orbital eccentricity to 0.01 and the orbital period to 4 days. The other parameters are equal to their reference values (see Table 3). Figs. 6a–6i display the lateral distributions of the maximum tidal dissipation rate in the interior (a, b, c), of the tidal dissipation rate at the surface (d, e, f), and of the surface heat flow (g, h, i). The latter value is computed by integrating the volumetric tidal heating over radial distance.

The maximum value of the radially dependent tidal dissipation rate is reached at $r = 1000 \text{ km}$ (a), and at the CMB (b, c) for undifferentiated and differentiated interiors, respectively. In any case, the dissipation patterns do not coincide with the tidal forcing potential patterns. The formers change significantly between the depth of the maximum dissipation rate and the surface, because they are controlled by the different components of the stress and strain tensors. Due to the liquid state of the core, the tangential stress y_4 vanishes at the CMB in the case of differentiated interiors, and the dissipation pattern is governed by the components in $\theta - \theta$,

$\theta - \phi$, $\phi - \phi$ directions [Eq. (10)]. For a solid homogeneous interior, the latter three components dominate progressively toward the surface, where the stress-free conditions lead to a dissipation distribution similar to the pattern observed at the CMB within differentiated interiors (compare Fig. 6d with Figs. 6b and 6c).

For a given mean density of the satellite, a lower value of the core density corresponds to an increase of the core radius, or equivalently to a thinning of the silicate mantle. Consequently, the tidal dissipation energy is concentrated into a thinner shell (see Fig. 3). This phenomenon leads to an increase of the dissipation rates with increasing core radius.

4.2.4. Influence of the outer H₂O layer

Let us now consider the effect of adding an outer H₂O layer on tidal dissipation in the deep interior.

4.2.5. Radial functions

Fig. 7 presents the radial functions for models with a liquid core ($\rho_C = 5150 \text{ kg m}^{-3}$) and different outer H₂O layers:

- a 120 km thick ice I layer (Model 1),
- a 119 km thick ocean overlain by a thin (1 km) ice I layer (Model 2),
- a high-pressure ice layer (600 km) overlain by an ocean (100 km) and by an ice I layer (100 km) (Model 3).

Models 1 and 2 can be considered as Europa-like models and Model 3 may resemble Titan's interior.

Adding an outer layer reduces the values of both radial and tangential stresses and displacements in the mantle, in particular when an ocean is incorporated in the model (Fig. 7). The reduction reaches a factor of two in the case of Model 3 (HP ice, ocean, and ice I). If ice I is directly in contact with the mantle (Model 1), ice exerts a traction at the top of the mantle (Fig. 7c). Conversely, if an ocean is present (Models 2 and 3), the outer ice I layer is decoupled from the

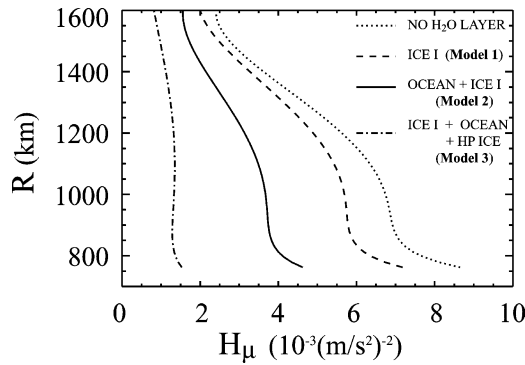


Fig. 8. Sensitivity parameter H_μ within the mantle for different models of the outer H_2O layer, and for a liquid core of 5150 kg m^{-3} .

deep interior, and the mantle undergoes compressive (negative) radial stress.

4.2.6. Radial and lateral distribution of the tidal dissipation rate

Fig. 8 shows the influence of the different outer shell structures on the sensitivity parameter H_μ in the mantle ($\rho_C = 5150 \text{ kg m}^{-3}$). Adding an outer shell reduces the value of the sensitivity parameter H_μ , and thus the value of the tidal dissipation rate. This reduction can reach a factor of four if part of the outer layer is liquid (Models 2 and 3). The shape of the H_μ profile varies little with different outer shell models. Only the mean value of H_μ is modified. The same remark holds for the lateral distribution of the tidal dissipation.

4.2.7. Summary

We have identified several parameters that influence, on one hand, the amplitude of the rocky interior response and, on the other hand, the distribution of the dissipation rate within it. The amplitude of the response is primarily controlled by the values of the silicate shear modulus μ_{sil} and of the viscosity η_{sil} , when it approaches the Maxwell value η_M . The core radius and the structure of the outer H_2O shell also play significant roles on the amplitude of the response. The presence of an ocean makes the deep interior less dissipative. The radial and lateral distributions of the dissipation rate strongly depend on the existence of a metallic core and on its density (or radius). The elastic bulk modulus K_{sil} and the orbital frequency ω have only minor effects.

4.3. Dissipation in the outer H_2O shell

In this section, we focus on the dissipation within ice layers for different outer layer models (Models 1, 2, 3, see above) and for a differentiated deep interior model ($\rho_C = 5150 \text{ kg m}^{-3}$). We explore systematically the parameters that have most influence on the amplitude and the distribution of the dissipation rate.

4.3.1. Radial distribution: sensitivity to shear anelasticity

Fig. 9 displays the sensitivity parameter H_μ within the H_2O layer for Titan-like model (Model 3). In the outer ice I

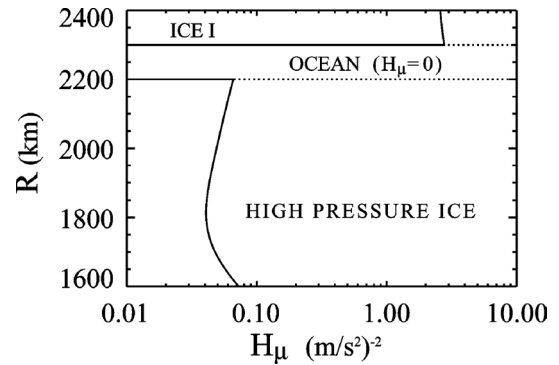


Fig. 9. Sensitivity parameter H_μ in the H_2O layer for a thick-layer Titan-like model composed of a high-pressure ice sublayer, an ocean and an ice I sublayer (Model 3).

sublayer above the ocean, the H_μ value remains almost constant with depth, whereas, in the high-pressure ice sublayer, it is more depth-sensitive because of the stress and displacement continuity between ice and the underlying solid mantle. The outermost ice I layer is at least one order of magnitude more sensitive to shear dissipation than the deep high-pressure ice. For Europa-like model with ocean (Model 2), the H_μ parameter in the ice I layer (not shown) is also constant with depth like in the outer layer in Model 3 (Fig. 9). The particular behavior of the H_μ parameter in the ice I layer is due to the radial displacement of the underlying ocean, which imposes high-amplitude tidal deformation at the base of the ice I layer.

4.3.2. Lateral distribution

The lateral distribution of the tidal dissipation rate at the bottom and at the top of the different types of ice layers are shown on Fig. 10. For these calculations, the reference values of the model parameters have been used (see Table 3), and the values of the eccentricity and of the orbital period have been prescribed to 0.01 and 4 days, respectively. For any ice layer overlying the silicate mantle (ice I layer in Model 1 or HP ice layer in Model 3), the lateral distribution of the dissipation rate as well as its mean value, dramatically change between the bottom and the top of the layer (Figs. 10a–10b, and Figs. 10e–10f). In Model 2, the distribution of the tidal dissipation rate remains constant throughout the ice I layer (Figs. 10c–10d), due to the decoupling effect of the underlying ocean. Like in the silicate mantle, the dissipation is dominated by the components in the θ – θ , θ – ϕ , and ϕ – ϕ directions of the stress-strain tensors. Note the high and rather constant value of the tidal dissipation rate in the outer ice I layer compared to that of the ice layers overlying the silicate mantle (Figs. 10a–10b, 10e–10f). As explained above, the dissipation difference is due to the presence of the ocean. For the example shown in Figs. 10c–10d, the volumetric tidal dissipation rate is about two orders of magnitude higher than the volumetric heating rate due to the radioactive decay in the silicate mantle.

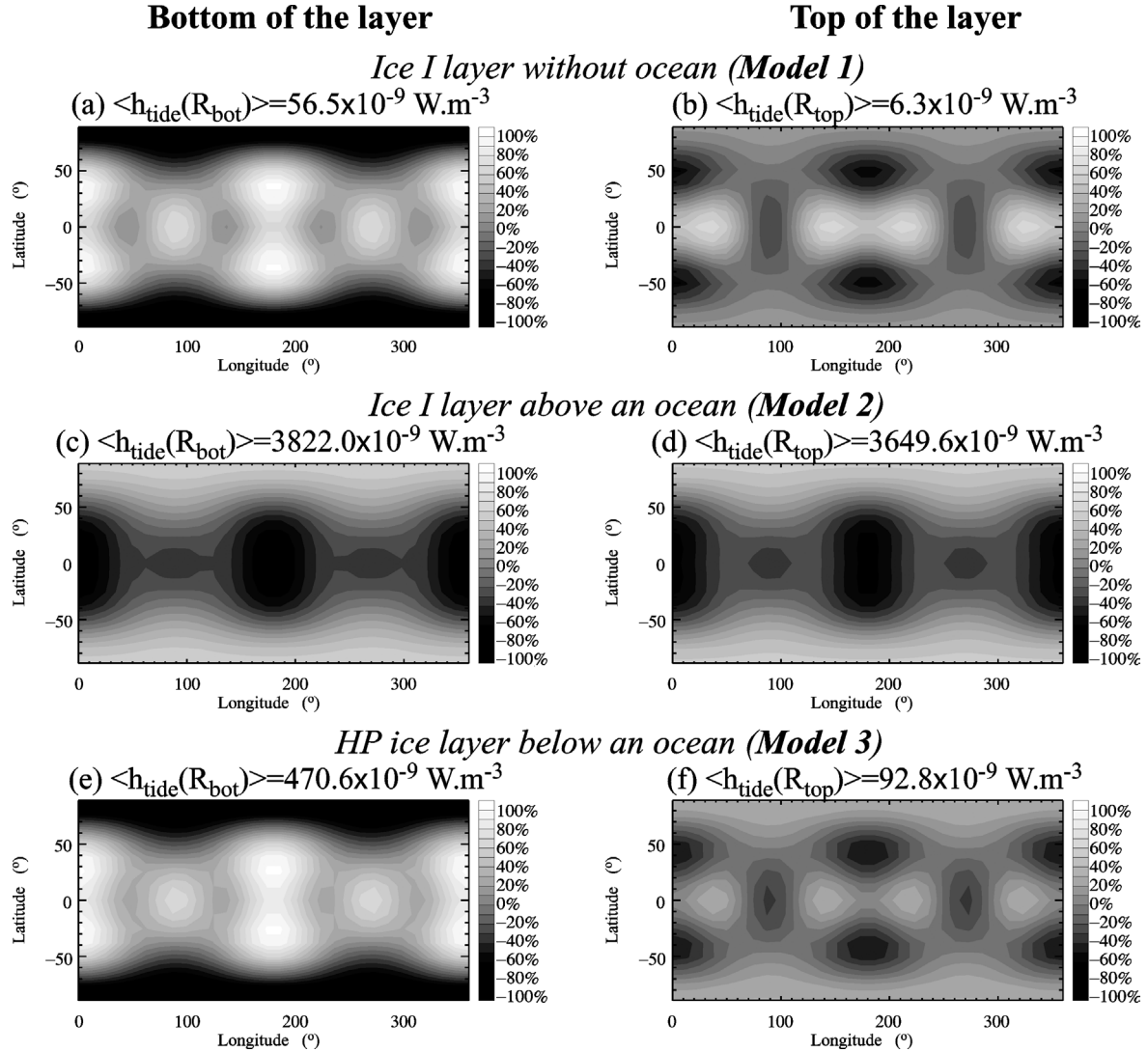


Fig. 10. Lateral distribution, in percent (%) of the mean value indicated above each panel, of the tidal dissipation rate at the bottom and at the top of the ice I layer of Model 1 (a, b), of the ice I layer above an ocean (c, d, Model 2), and of the high-pressure ice layer of Model 3 (e, f).

4.3.3. Influence of the ice parameters

The viscosity and shear modulus dependences of the $\langle H_{\mu} \rangle$ value averaged throughout the ice layers are shown on Fig. 11. The latter value depends very little on the parameters in the ice I shell above the liquid layer (Fig. 11(2)). Conversely, it is very sensitive to the value of the shear modulus and particularly to that of the viscosity in the ice shells coupled with the mantle (Figs. 11(1) and 11(3)).

If an ocean is present, both with and without HP ice layer (Model 2 or Model 3), the variations of the parameters (η_1 , μ_1 , and K_1) in the outer ice I layer do not significantly modify the radial and lateral distribution of the dissipation rate. For any ice layer overlying the silicate mantle, the radial and lateral distribution are sensitive to the variations of the values of both the shear modulus and the viscosity, particularly at the base of the layer. The decrease of the shear modulus or of the viscosity tends to increase the lateral variations of the tidal dissipation rate.

4.3.4. Influence of the deep interior structure

The structure of the deep interior has an influence only if the ice layer is coupled with it. It particularly modifies the response at the base of the ice layer. For the ice layer above the ocean, the tidal response is mainly sensitive to the relative thickness of the ocean compared to that of the ice layer. It can be shown from Eq. (33) that, due to the absence of shear stress in the ocean, and to the boundary conditions at the discontinuous interface between the ocean and the overlying ice I layer, the mean value of H_{μ} in the latter layer is proportional to the square of the radial displacement (y_1) induced by the ocean.

4.3.5. Summary

Our calculations show that both the amplitude and the distribution of the tidal dissipation rate in the outer layer strongly depend on the presence of an ocean. The ocean, by inducing a large radial displacement, imposes the deforma-

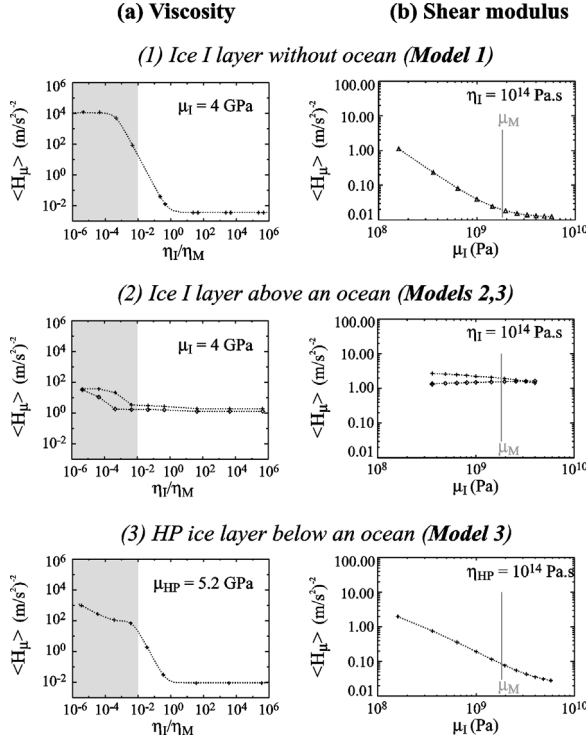


Fig. 11. Sensitivity parameter $\langle H_\mu \rangle$ averaged throughout the whole ice layer as a function of viscosity (a) and of shear modulus (b), for the different layer composing Models 1, 2, and 3. Like for Fig. 5b, the ratio between the velocities V_p and V_s is prescribed to $\sqrt{3}$, when V_s , i.e. shear modulus μ , varies. The gray area on plots (a) indicates the viscosity values that are physically unreliable.

tion in the overlying ice layer, so that the response depends much less on the parameters of the layer and on the deep internal structure. Conversely, if the ice layer is coupled with the silicate mantle, several parameters (core density and radius, interior state, rigidity and viscosity of the mantle and of the ice layer) must be taken into account to describe its response to the tide-raising potential.

5. Discussions and conclusion

Using a new method of tidal dissipation calculations relating the tidal dissipation rate to the radial distribution of the rigidity, we have characterized the tidal dissipation within a generic satellite displaying Io-, Europa-, and Titan-like internal structures. By exploring a series of interior parameters and by progressively adding outer H_2O layers to an initial iron and silicate interior similar to Io's, we have demonstrated the strong influence of a liquid water layer on the distribution of dissipation within both the deep interior of the satellite and the outer ice I layer. The advantage of the generic satellite approach, although it does not reproduce the dissipation for each individual satellite, is to identify the influence of each internal layer on the distribution of dissipation, and to highlight the fundamental difference in the tidal dissipation distribution between Io, Europa and the larger icy satellite, like Titan or Ganymede.

The liquid layer, by decoupling the deep interior from the surface reduces the strength of dissipation in the inner part of the satellite and increases it in the outer part above the ocean. It totally controls the lateral distribution within the outer ice I layer. Several previous studies (e.g., Cassen et al., 1979; Ojakangas and Stevenson, 1989; Sohl et al., 1995; Castillo et al., 2000; Moore and Schubert, 2000; Wu et al., 2001; Moore and Schubert, 2003; Sohl et al., 2003) have already pointed out the primary effect of the subsurface ocean on the tidal response of the satellite and on the global dissipation in the outer ice I layer. Calculations done in the present study indicate more precisely that the tidal deformation of the outer ice I layer, which is proportional to the H_μ parameters, is mainly controlled by the radial displacement of the subsurface ocean. The tidal strain rate is almost constant with depth, and it is maximum at the poles and minimum at the equator. We have shown that these characteristics are not sensitive to variations of ice rheology parameters within the ice I layer. Consequently, lateral variations of viscosity related to convective instabilities in the outer ice I layer are not expected to modify significantly the tidal strain rate field within the layer. Following this, the dissipation rate is directly related to the viscosity value through the viscosity dependence of the imaginary part of the shear modulus $\text{Im}\{\mu\}$ [Eqs. (13) and (37)].

Conversely, in the cases of large icy satellite where high-pressure ice layers are present below the liquid layer, a simple scaling cannot be determined for two reasons. First, the tidal dissipation rate is a function of viscosity through the viscosity dependence of both the H_μ parameter and the imaginary part of the complex shear modulus $\text{Im}\{\mu\}$. Second, the lateral distribution of dissipation is highly sensitive to depth within the layer (Fig. 10). However, the volumetrically tidal dissipation is at least one order of magnitude smaller in the high-pressure ice layer than in the outer ice I layer, for the same value of viscosity close to 10^{14} Pa.s. Due to the smaller dissipation in the high-pressure layer, the effect of lateral and radial distribution on the layer thermal state can be neglected at first order approximation, whereas it cannot be within the outer ice I layer.

Assuming the same constitutive parameters (reference parameters in Table 3) for Europa and Io and interior models ($\rho_C = 7000 \text{ kg m}^{-3}$, $\rho_M = 3300 \text{ kg m}^{-3}$) consistent with the Galileo gravimetric data (Sohl et al., 2002), our calculation method shows that the computed H_μ parameter within Europa's mantle is at least three times lower than within Io's mantle, due to the decoupling effect of Europa's ocean and due to its smaller size (Fig. 12). By lowering the dissipation rate in the mantle, the presence of a subsurface ocean within Europa limits the impact of tidal dissipation on its interior thermal evolution. This result suggests that the common scaling used to estimate the upper bound of tidal heating within Europa's interior from Io's surface heat flow (Thomson and Delaney, 2001) probably overestimates it. Nevertheless, the issue of whether tidal dissipation in the silicate mantle of Europa might have affected its thermal

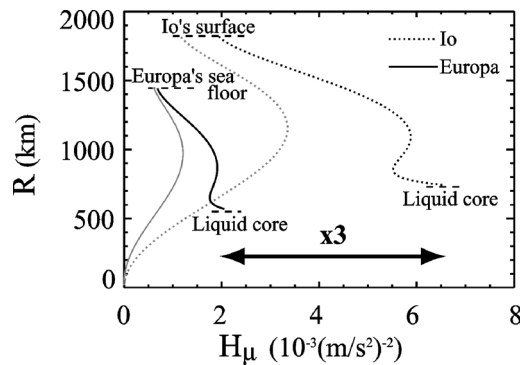


Fig. 12. Sensitivity parameter H_μ within Europa's and Io's silicate interior, before differentiation ($\rho = 3500 \text{ kg m}^{-3}$, gray curves) and after differentiation into a silicate mantle and a liquid iron core ($\rho_C = 7000 \text{ kg m}^{-3}$ and $\rho_M = 3300 \text{ kg m}^{-3}$, black curves).

history cannot be understood without considering the exact distribution of tidal dissipation, the influence of each internal layer (e.g., iron core, subsurface ocean) as well as the evolutions of its neighboring moons, Io and Ganymede. The new method presented in this paper provides a useful tool to study the coupled thermal-orbital evolution of the satellites (e.g., [Hussmann and Spohn, 2004](#)). It provides a self-consistent way to compute the distribution of tidal heating within the interior as the temperature profile evolves and as the liquid layer crystallizes with time [see [Tobie et al. \(2005\)](#) for an application to Titan's evolution].

Although we have assumed homogeneous internal layers and Maxwell rheology for the calculations presented in this paper, our calculation method can be applied to radially heterogeneous layers assuming any kind of rheological model. Assuming any other rheology for the interior would not modify our conclusions about the influence of the ocean on the tidal dissipation distribution, but it may change the viscosity dependence of the dissipation rate. Lateral heterogeneities associated with convective instabilities in the interior or thickness variations of the outer ice I layer ([Ojakangas and Stevenson, 1989](#); [Tobie et al., 2003](#)) may make the hypothesis of spherical symmetry not valid. In this case, other more complex theories taking into account the coupling between neighboring modes of deformation must be used (e.g., [Martinec, 2000](#)). However, as long as lateral variations of viscosities do not affect the tidal strain field, as for the outer ice I layer above an ocean, the three-dimensional distribution of tidal strain rate and the associated dissipation rate can be determined from the radial function y_i of the satellite and from the three-dimensional distribution of the imaginary part of the shear modulus, thus enabling the coupling of tidal heating with heat transfer process in a self-consistent way within any planetary interior.

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